

PLYMOUTH MOUNT BATTEN, United Kingdom

WMO: 038270

Lat: 50.3550N

Lon: 4.1211W

Elev: 89

StdP: 14.65

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.8	32.0	21.9	16.5	33.5	24.9	19.1	35.0	42.9	49.3	38.4	47.8	7.5	70	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	74.4	64.2	71.3	62.8	68.8	61.7	65.7	71.5	64.3	69.2	63.0	67.0	10.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.7	88.5	67.2	62.5	84.9	65.6	61.5	81.9	64.5	30.5	71.8	29.4	69.4	28.6	67.1	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
33.9	29.5	25.7	DB	26.1	79.6	2.6	3.5	24.2	82.1	22.7	84.1	21.2	86.1	19.3	88.6
			WB	25.0	69.0	2.9	2.9	22.9	71.1	21.2	72.8	19.6	74.5	17.5	76.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	43.9	43.7	45.7	49.0	53.6	58.4	61.7	61.8	58.8	54.3	48.6	45.1	
	DBStd	7.89	5.10	4.64	4.25	4.25	4.24	3.79	3.53	3.33	3.29	4.14	4.87	5.43	
	HDD50	843	191	177	140	65	12	0	0	0	0	12	82	164	
	HDD65	4739	653	596	597	481	355	204	115	109	188	331	493	617	
	CDD50	1612	3	2	8	35	123	251	363	367	263	146	39	12	
	CDD65	30	0	0	0	0	0	5	13	11	1	0	0	0	
	CDH74	115	0	0	0	0	1	29	47	36	2	0	0	0	
	CDH80	9	0	0	0	0	0	2	2	5	0	0	0	0	
Wind	WSAvg	11.7	14.0	13.1	12.4	11.0	10.9	10.4	10.0	9.9	10.4	12.4	12.4	13.6	
Precipitation	PrecAvg	39.5	4.4	3.6	3.3	2.4	2.4	2.2	2.2	2.7	3.1	3.9	4.3	4.7	
	PrecMax	52.5	8.5	8.1	7.0	6.0	4.9	5.7	5.2	5.6	7.2	8.3	8.9	9.0	
	PrecMin	29.1	0.5	0.1	0.1	0.2	0.2	0.2	0.3	0.2	0.5	0.3	1.6	0.9	
	PrecStd	5.8	2.2	2.1	1.6	1.3	1.4	1.5	1.2	1.5	1.8	1.9	1.9	2.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	54.2	59.8	67.3	71.7	78.3	78.9	78.8	72.4	66.1	59.1	55.8	
		MCWB	52.2	50.3	51.9	57.1	61.9	64.6	65.8	66.9	63.2	60.8	56.9	54.2	
	2%	DB	52.7	52.2	55.5	61.3	67.0	71.8	74.5	72.9	68.8	62.8	57.5	54.7	
		MCWB	51.4	50.0	50.0	53.6	58.6	61.6	64.1	64.4	61.9	58.4	55.7	53.4	
	5%	DB	51.8	51.1	53.0	57.7	63.6	68.0	71.4	69.8	66.3	61.1	56.5	53.8	
		MCWB	50.6	49.3	49.1	51.4	57.1	60.0	63.2	62.9	60.4	58.0	54.8	52.5	
	10%	DB	50.9	50.1	51.4	55.3	60.8	65.3	68.5	67.5	64.4	59.9	55.5	52.8	
		MCWB	49.5	48.4	48.5	50.2	55.2	58.8	61.9	61.9	59.2	57.4	53.9	51.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	52.1	53.0	58.0	63.0	66.3	67.4	69.3	65.2	61.9	57.9	54.9
			MCDB	53.2	53.3	56.6	65.6	68.9	75.3	75.3	74.7	69.4	64.5	58.3	55.2
2%		WB	51.9	51.1	51.5	54.3	60.0	63.0	65.4	65.9	63.3	60.2	56.5	54.0	
		MCDB	52.3	51.7	53.6	59.3	65.2	69.6	72.0	70.5	66.5	61.5	57.0	54.4	
5%		WB	51.0	50.0	50.4	52.6	57.9	61.3	64.1	64.4	62.1	59.2	55.5	53.1	
		MCDB	51.6	50.7	52.0	56.2	62.3	66.5	69.8	68.1	64.5	60.3	56.2	53.6	
10%		WB	49.9	48.9	49.5	51.3	56.1	59.8	62.8	63.1	61.0	58.2	54.3	51.8	
		MCDB	50.7	49.9	50.9	54.2	59.7	63.9	67.1	66.1	63.0	59.4	55.3	52.6	
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.7	9.8	11.5	11.9	11.9	11.7	11.7	11.5	9.4	8.7	7.9	
		MCDBR	7.0	8.7	11.9	15.0	15.7	16.7	16.6	14.6	13.9	10.2	7.8	6.8	
		MCWBR	7.6	7.4	8.3	9.6	9.9	9.8	9.0	8.4	8.5	7.3	7.1	7.4	
	5% WB	MCDBR	6.9	7.7	9.6	12.4	14.0	14.3	14.1	12.3	11.4	8.3	6.8	6.5	
		MCWBR	7.9	7.6	7.8	8.7	9.5	9.1	8.6	8.0	8.0	7.0	7.0	7.7	
Clear-Sky Solar Irradiance	taub		0.335	0.349	0.379	0.390	0.396	0.385	0.385	0.372	0.367	0.359	0.333	0.326	
	taud		2.352	2.371	2.297	2.293	2.318	2.381	2.387	2.440	2.441	2.431	2.407	2.321	
	Ebn at Noon		217	247	259	270	273	277	275	273	262	241	218	200	
	Edh at Noon		21	27	34	38	38	36	36	32	29	25	20	19	
All-Sky Solar Radiation	RadAvg		259	506	880	1408	1703	1864	1732	1470	1109	626	330	207	
	RadStd		23	51	101	104	108	174	120	105	73	43	29	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page